Board Manual

for Development Kit 4

MMSP™2

User Manual

MDMP2520FDEB4UM

Ver 1.0 / June 20, 2005

MP2520F

Board Manual for Development Board based MMSP 2 Processor

Available Board - Development Board 4.1.1

MAGIC EYES

Important Notice

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1. Introduce to Development Board 4.1.1

1.1. Specification

● MagicEyes MP2520F Application Processor

● LCD Display Components

-7 inch TFT-LCD

- WVGA(800 x 480) 260K Color

● Memory

- SDRAM : 64MB (Up to 128MB)

- NAND Flash 64MB

● Storage

- MMC, SD(Sub Board)
- Compact Flash
- 2.5, 1.8 inch HDD Interface

● USB

- Host : USB 1.1 : 1 ch
- Device : USB 2.0 1 ch, Using Netchip 2272.
  : USB 1.1 1ch, Direct connect to MMSP2

● UART

- Half UART 1ch
- Baud rate : 9,600 ~ 115,200bps

● Digital camera Module

- CMOS Image sensor : Support only Omnivision, OV9640
- Camera Module : Support only Nanovision, NVM-O9640
  - 1.3M Pixel, 1/4 inch, CMOS
  - Progressive Scan
  - YUV 8 bit output
  - View Angle : 36(V) / 47(H) / 60(D)
  - Pixel : 1280 X 960

● Video

- Video out (CVBS, A-RGB, S-Video, YPbPr)
- Video In (CVBS, S-Video)
- D-SUB for Monitor

● Audio

- Stereo out x 1
- MIC In x 2
- Line In x 1

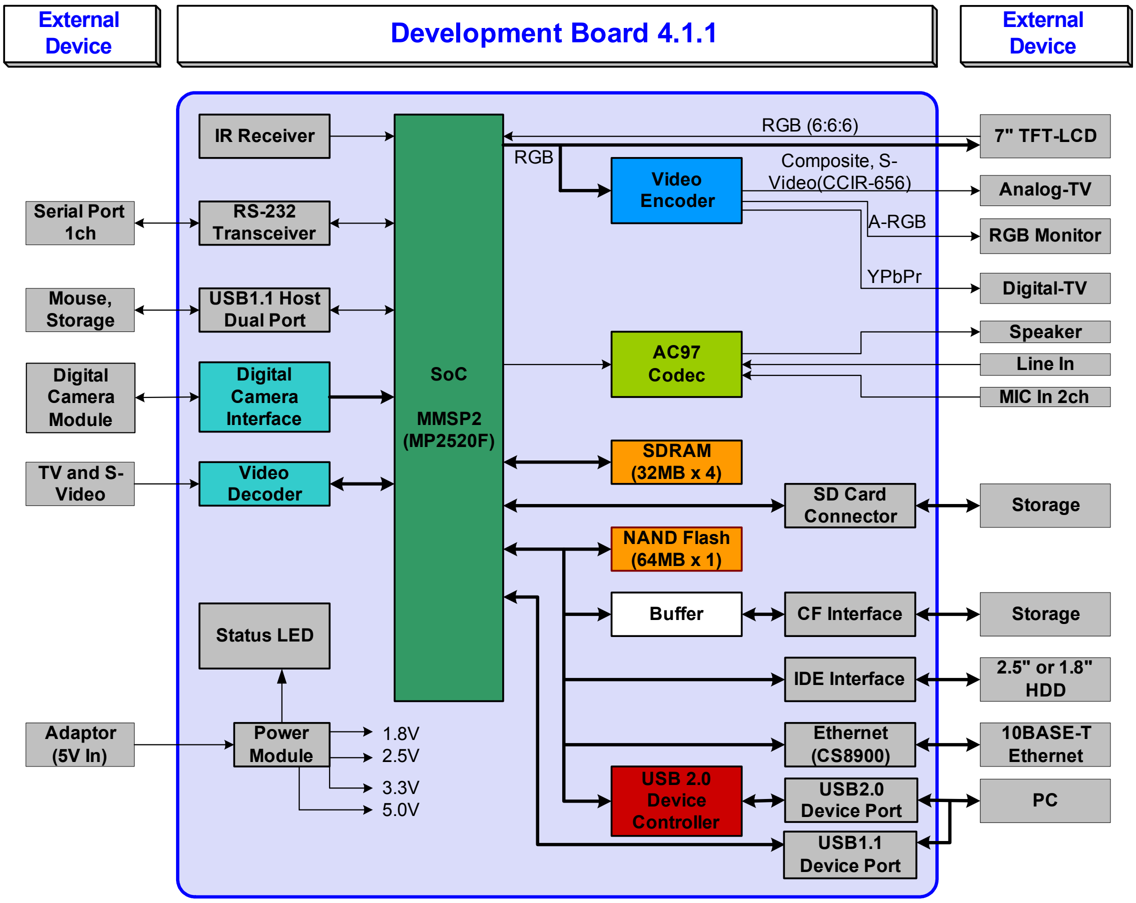
● Network (Sub Board)

- Ethernet 10 BaseT x 1

● JTAG Port

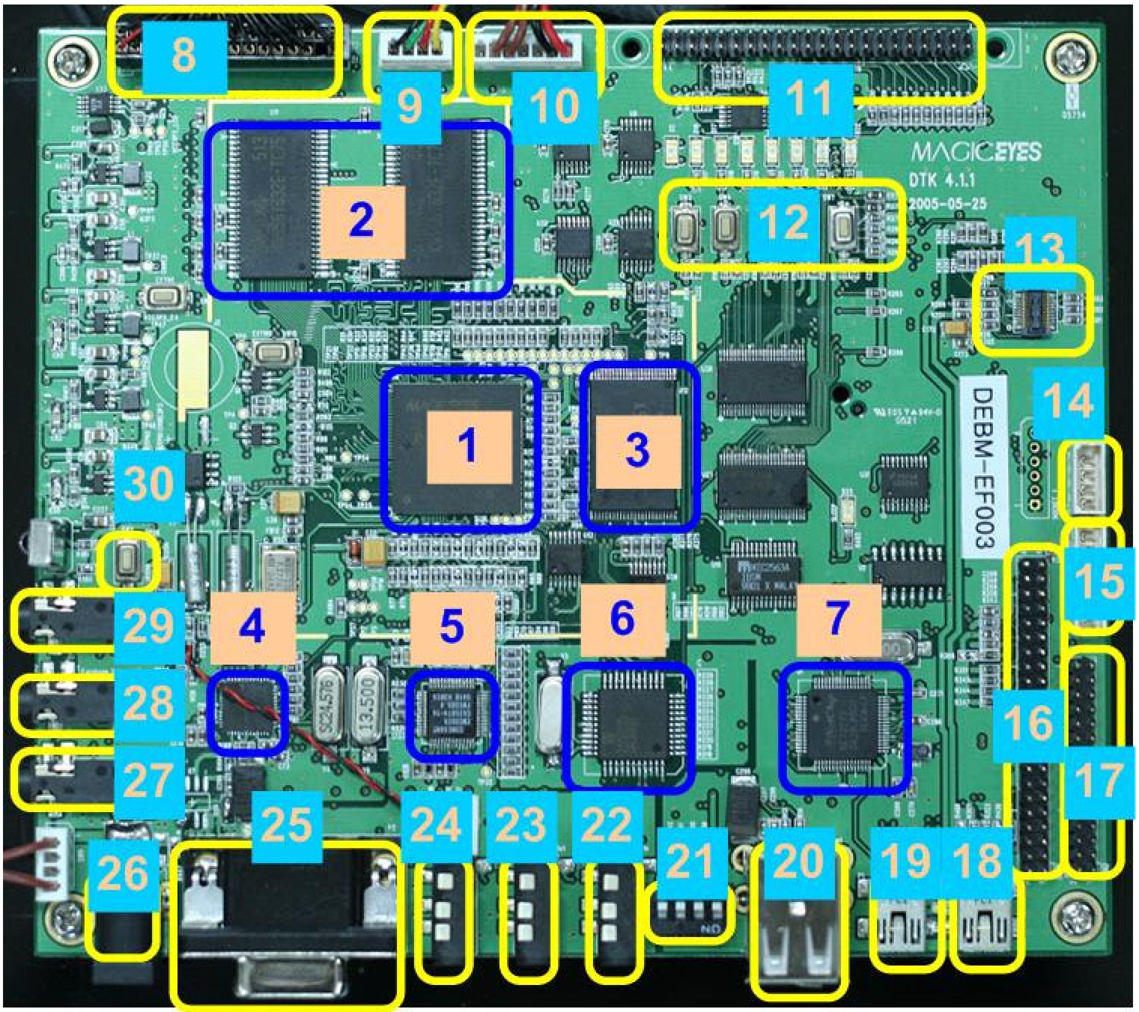
● Power Supply : 5V/3A Adaptor

1.2. Block Diagram



<Fig 1. 1. Block Diagram>

1.3. Parts arrangement



<Fig 1. 2. Parts Arrangement>

No	Description	Remark	Manufacture
1	CPU	MP2520F	MagicEyes
2	SDRAM	K4S561632E-TC75	Samsung
3	NAND Flash	K9F1208U0M	Samsung
4	Audio Codec	WM9710	Wolfson
5	Video Encoder	CX25874	Conexant
6	Video Decoder	SAA7113H	Philips
7	USB 2.0 Device Controller	NET2272	Netchip

8	LCD Connector		
9	Touch Connector		
10	Inverter Connector		
11	IDE Connector		
12	Key pad	5ea	
13	Camera Connector	OVM9640	Nanovision
14	UART Connector		
15	JTAG Connector		
16	SDMMC Board Connector	SD/MMC socket	
17	Ethernet Board Connector	CS8900A	Cirrus_Logic
18	USB B Type Connector	USB 1.1 Device	For MMSP2 (USB 1.1)
19	USB B Type Connector	USB 2.0 Device	For Net2272 (USB 2.0)
20	USB A Type	USB 1.1 Host	Two Channel
21	Dip Switch	Display Setting	Use only WinCE
22	Video In Jack	CVBS, S-Video	
23	Video Out Jack	CVBS, S-Video	
24	Video Out Jack	YPbPr	
25	Video Out Connector	RGB	
26	Power DC Jack	5V/3A	
27	Audio In Jack	Line In	
28	Audio Out Jack	Stero Out	
29	Audio In Jack	Mic In	
30	Button	Reset Button	

<Table 1. 1. Parts Description>

2. Device Description

2.1. LCD

- 7” WVGA Resolution(800x480)
- Color depth : 6bit, 262,144color
- Power consumption : 2.275W(Typ)

2.1.1. LCD Signals

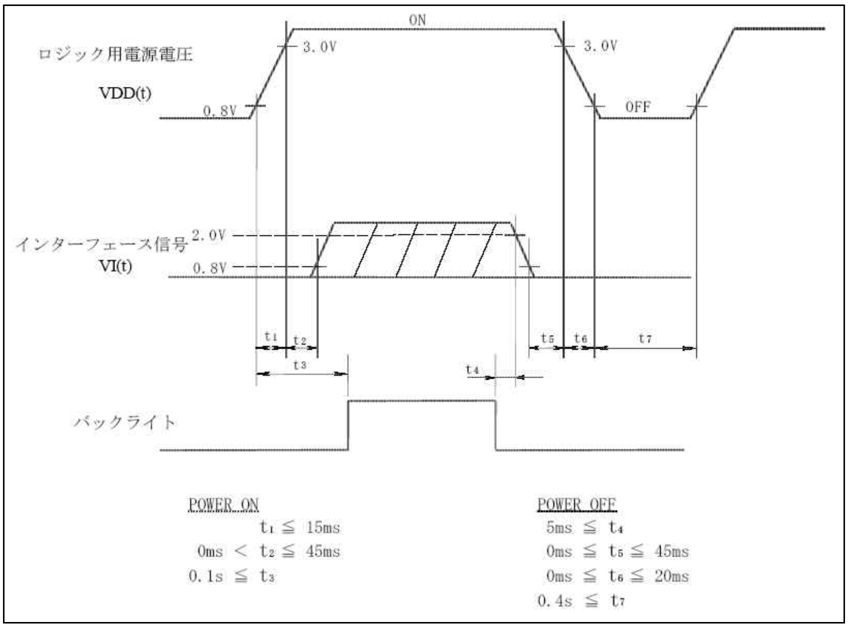
Pin	Net Name	Type	Configuration	Description
GPIOB[13:8]	VD[5:0]	Alt1	NONE	RED[5:0]
GPIOB[15:14] GPIOA[3:0]	VD[11:6]	Alt1	NONE	GREEN[5:0]
GPIOA[9:4]	VD[17:12]	Alt1	NONE	BLUE[5:0]
GPIOB[4]	VSYNC	Alt1	NONE	Vertical SYNC
GPIOB[5]	HSYNC	Alt1	NONE	Horizontal SYNC
GPIOB[6]	ACTIVE	Alt1	NONE	Data Enable
GPIOB[7]	CLKHO	Alt1	NONE	LCD Clock

<Table 2. 1. GPIO for LCD Signals>

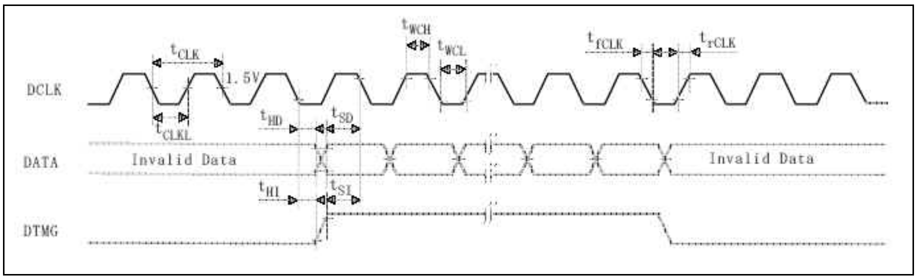
2.1.2. LCD Backlight & Brightness Control

Pin	Net Name	Type	Configuration	Description
GPION[0]	BL_ON_OFF	Output	Ext P/D	Back Light ON/OFF control
GPIOL[11]	LCD_BR_CON	Alt1	NONE	LCD brightness control by PWM

<Table 2. 2. GPIO for LCD Backlight>



<Fig 2. 1. LCD Power On Sequence>



<Fig 2. 2. Signal Timing Waveform>

2.2. NAND Flash Memory (K9F1208U0M-YCB0)

- CS : nNFCE[0]
- BUS width : 8bit
- NAND Boot size : 512byte
- NAND flash type : 4 address
- DTK4.1.1 supports Large block NAND.

2.2.1. Data Bus (8bit)

Pin	Net Name	Type	Configuration	Description
GPIOE[0]	ZD0	Alt1	NONE	
GPIOE[1]	ZD1	Alt1	NONE	
GPIOE[2]	ZD2	Alt1	NONE	
GPIOE[3]	ZD3	Alt1	NONE	
GPIOE[4]	ZD4	Alt1	NONE	
GPIOE[5]	ZD5	Alt1	NONE	
GPIOE[6]	ZD6	Alt1	NONE	
GPIOE[7]	ZD7	Alt1	NONE	

<Table 2. 3. GPIO for NAND Flash Data BUS>

Pin	Name	Type	Configuration	Description
GPIOJ[0]	nNFWE	Alt1	NONE	Write Enable
GPIOJ[1]	nNFOE	Alt1	NONE	Read Enable
GPIOJ[2]	F_RnB	Alt1	NONE	NAND Ready/nBUSY signal
GPIOI[0]	nNFCS0	Alt1	Ext P/U	NAND Flash Chip Enable
GPIOJ[10]	nPIOR	Alt1	NONE	NAND Flash Command Latch Enable
GPIOJ[9]	nPIOW	Alt1	NONE	NAND Flash Address Latch Enable
GPIOI[2]	nNFWP	Output	Ext P/U	NAND Flash Write Protect 0 : Write Protect Enable 1 : Write Protect Disable
GPIOO[1]	GPIOO_1	Output	Ext P/D	0 → 1 : Large Block Nand 1 : Small Block Nand

<Table 2. 4. GPIO for NAND Flash Control BUS>

2.3. UART

2.3.1. UART for Debug

● Hardware flow control

Pin	Net Name	Type	Configuration	Description
GPIOD[0]	Tx0	Alt1	NONE	UART0 TX (Output)
GPIOD[1]	Rx0	Alt1	NONE	UART0 RX (Input)

<Table 2. 5. GPIO for UART>

2.4. Buttons and Dip Switch

2.4.1. Buttons

Pin	Net Name	Type	Configuration	Description
GPIOC[11]	KEY_ENTER	Input	Ext P/U	
GPIOC[12]	KEY_EXIT	Input	Ext P/U	
GPIOC[13]	KEY_MENU	Input	Ext P/U	
GPIOC[14]	KEY_DOWN	Input	Ext P/U	
GPIOC[15]	KEY_UP	Input	Ext P/U	

<Table 2. 6. GPIO for Key buttons>

2.4.2. DIP Switch

Pin	Net Name	Type	Configuration	Description
GPIOG[11]	GPIOG_11	Input	Ext P/U	
GPIOG[12]	GPIOG_12	Input	Ext P/U	
GPIOG[13]	GPIOG_13	Input	Ext P/U	
GPIOG[14]	GPIOG_14	Input	Ext P/U	

<Table 2. 7. GPIO for DIP Switch>

● DIP switch is used for [Display Mode Setting] for WinCE

● Dip Switch Set Value

Development Board 4	SW1	4	3	2	1
-	-	OFF	OFF	OFF	OFF
D-SUB(Monitor)	640x480x60Hz	OFF	OFF	OFF	ON
D-SUB(Monitor)	800x600x60Hz	OFF	OFF	ON	OFF
Component(YPbPr)	720x480x60Hz	OFF	OFF	ON	ON
DVI	800x600x60Hz	OFF	ON	OFF	OFF
Composite/ S-Video	720x480x60Hz	OFF	ON	OFF	ON
LCD	800x600x60Hz	OFF	ON	ON	OFF
LCD	640x480x60Hz	OFF	ON	ON	ON

<Table 2. 8. Development Board 4.1.1 Display Mode Setting Switch >

<Note>

- **Development Board 4.1.1 does not support DVI output.**

2.5. SD Card

Pin	Name	Type	Configuration	Description
GPIOL[0]	SD_DAT0	Alt1	Ext P/U	SD Data0
GPIOL[1]	SD_DAT1	Alt1	Ext P/U	SD Data1
GPIOL[2]	SD_DAT2	Alt1	Ext P/U	SD Data2
GPIOL[3]	SD_DAT3	Alt1	Ext P/U	SD Data3
GPIOL[4]	SD_CMD	Alt1	Ext P/U	SD Command
GPIOL[5]	SD_CLK	Alt1	NONE	SD Clock
GPIOD[2]	SD_WP	Input	Ext P/U	SD Write Protect
GPIOI[14]	SD_CD	Input	Ext P/U	SD Card Detect

<Table 2. 9. GPIO for SD Card>

2.6. Audio

Pin	Name	Type	Configuration	Description
GPIOL[6]	AC97_DATOUT	Alt1	NONE	Audio Data Out
GPIOL[7]	AC97_nRST	Alt1	Ext P/U	0 : Reset 1 : Normal
GPIOL[8]	AC97_SYNC	Alt1	NONE	Audio Sync
GPIOL[9]	AC97_DATIN	Alt1	NONE	Audio Data In
GPIOL[10]	AC97_BITCLK	Alt1	NONE	Audio Bit Clock

<Table 2. 10. GPIO for Audio>

- CODEC : Wolfson WM9710
- DTK 4.1.1 has 2port Mic In

2.6.1. Analog Path

- MICIN : Microphone Input 1, 2
- LINEIN_L, LINEIN_R : Audio Input
- HPOUT_L, HPOUT_R : Headphone Output
- Variable sample rates are supported

Audio Sample Rate	Control Value (D15 – D0)
8000	1F40
11025	2B11
16000	3E80
22050	5622
32000	7D000
44100	AC44
48000	BB80

<Table 2. 11. Audio Sample Rate>

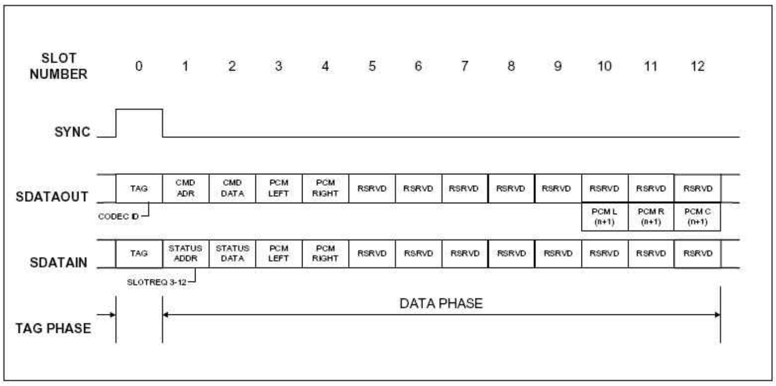
2.6.2. Volume Control

- Stereo Master Volume : 02h
- Stereo Headphone Volume : 04h
- Mono Volume : 06h

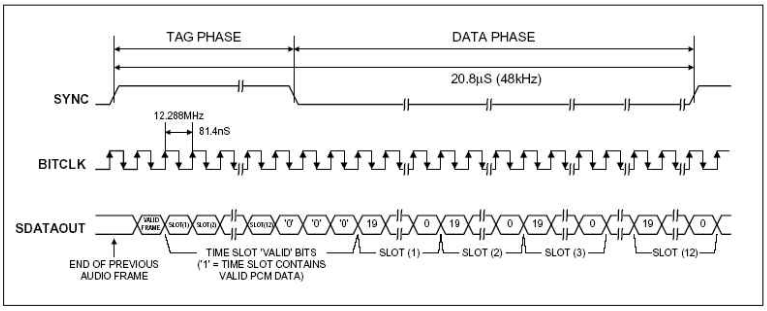
Reg	Name	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Default
00h	Reset	X	SE4	SE3	SE2	SE1	SE0	ID9	ID8	ID7	ID6	ID5	ID4	ID3	ID2	ID1	ID0	6150h
02h	Master volume	Mute	X	X	ML4	ML3	ML2	ML1	ML0	X	X	X	MR4	MR3	MR2	MR1	MR0	8000h
04h	HPHONE volume	Mute	X	X	ML4	ML3	ML2	ML1	ML0	X	X	X	MR4	MR3	MR2	MR1	MR0	8000h
06h	Master volume mono	Mute	X	X	X	X	X	X	X	X	X	X	MM4	MM3	MM2	MM1	MM0	8000h
0Ah	PCBEEP volume	Mute	X	X	X	X	X	X	X	X	X	X	PV3	PV2	PV2	PV0	X	8000h
0Ch	Phone volume	Mute	X	X	X	X	X	X	X	X	X	X	GN4	GN3	GN2	GN1	GN0	8008h
0Eh	Mic volume	Mute	X	X	X	X	X	X	X	X	20dB	X	GN4	GN3	GN2	GN1	GN0	8008h
10h	Line in volume	Mute	X	X	GL4	GL3	GL2	GL1	GL0	X	X	X	GR4	GR3	GR2	GR1	GR0	8808h
12h	CD volume	Mute	X	X	GL4	GL3	GL2	GL1	GL0	X	X	X	GR4	GR3	GR2	GR1	GR0	8808h
18h	PCM out volume	Mute	X	X	GL4	GL3	GL2	GL1	GL0	X	X	X	GR4	GR3	GR2	GR1	GR0	8808h
1Ah	Rec select	X	X	X	X	X	SL2	SL1	SL0	X	X	X	X	X	SR2	SR1	SR0	0000h
1Ch	Rec gain	Mute	X	X	X	GL3	GL2	GL1	GL0	X	X	X	X	GR3	GR2	GR1	GR0	8000h
20h	General purpose	POP	X	3D	X	X	X	MIX	MS	LPBK	X	X	X	X	X	X	X	0000h
22h	3D control	X	X	X	X	X	X	X	X	X	X	X	X	DP3	DP2	DP1	DP0	0000h
26h	Power/down control status	APD	PR6	PR5	PR4	PR3	PR2	PR1	PR0	X	X	X	X	REF	ANL	DAC	ADC	000Fh
28h	Ext'd audio ID	ID1	ID0	X	X	REV1	REV0	AMAP	LDAC	SDAC	CDAC	DSA1	DSA0	VRM	SPDIF	DRA	VRA	0605h
2Ah	Ext'd audio stat/ctrl	X	X	X	X	X	SPCV	X	X	X	X	SPSA1	SPSA0	X	SPDIF	X	VRA	0000h
2Ch	Audio DAC rate	SR15	SR14	SR13	SR12	SR11	SR10	SR9	SR8	SR7	SR6	SR5	SR4	SR3	SR2	SR1	SR0	BB80h
32h	Audio ADC rate	SR15	SR14	SR13	SR12	SR11	SR10	SR9	SR8	SR7	SR6	SR5	SR4	SR3	SR2	SR1	SR0	BB80h
3Ah	SPDIF control	V	0	1	0	L	CC6	CC5	CC4	CC3	CC2	CC1	CC0	PRE	COPY	UDIB	PRO	2000h
5Ah	Mixer Path Mute	0	0	0	0	0	0	0	0	0	0	0	MPM	0	0	0	0	0000h
5Ch	Add. Function control	AMUTE	HSCP	PUEN	MHPZ	PSEL	HSEN	HSDT	HPND	AMEN	I²S	ADCN DAC	ADCO	HPF	HS CMP	ASS1	ASS0	0000h
72h	Front mixer volume	Mute	X	X	GL4	GL3	GL2	GL1	GL0	X	X	X	GR4	GR3	GR2	GR1	GR0	0808h
74h	Add. Function	X	X	X	X	X	X	X	X	X	X	X	X	HPB	I2S64	X	MONOEN	0000h
78h	Add. Function	0	0	0	0	0	0	0	0	0	PHIZ	0	0	0	0	0	0	0000h
7Ch	Vendor ID1	F7	F6	F5	F4	F3	F2	F1	F0	S7	S6	S5	S4	S3	S2	S1	S0	574Dh
7Eh	Vendor ID2	T7	T6	T5	T4	T3	T2	T1	T0	Rev7	Rev6	Rev5	Rev4	Rev3	Rev2	Rev1	Rev0	4C05h

<Table 2. 12. Serial Interface Register MAP>

2.6.3. AC-Link Serial Interface Protocol



<Fig 2. 3. AC97 Standard Bi-Directional Audio Frame>



<Fig 2. 4. AC-Link Audio Output Frame>

2.7. SDRAM (K4S561632C-TC/L1H)

- Samsung K4S561632E-TC75
- CS : nSCSx0, nSCSx1
- 256Mb(32MB), 4bank SDRAM x 2 (Up to x 4)
- 4M x 16Bit x 4 Banks Synchronous DRAM
- JEDEC standard 3.3V power supply
- LVTTL compatible with multiplexed address
- MRS cycle with address key programs
  - CAS latency (2 & 3)
  - Burst length (1, 2, 4, 8 & Full page)
  - Burst type (Sequential & Interleave)
- All inputs are sampled at the positive rising edge of the system clock.
- Burst read single-bit write operation
- DQM for masking
- Auto & self refresh
- 64ms refresh period (8K Cycle)
- Simplified Truth Table

Command			CKEn-1	CKEn	$\overline{CS}$	$\overline{RAS}$	$\overline{CAS}$	$\overline{WE}$	DQM	$BA_{0,1}$	A_{10}/AP	A_{11},A_{12}, A_9 ~ A_0	Note
Register	Mode register set		H	X	L	L	L	L	X	OP code			1,2
Refresh	Auto refresh		H	H	L	L	L	H	X	X			3
	Self refresh	Entry		L									3
		Exit	L	H	L	H	H	H	X	X			3
					H	X	X	X					3
Bank active & row addr.			H	X	L	L	H	H	X	V	Row address		
Read & column address	Auto precharge disable		H	X	L	H	L	H	X	V	L	Column address (A_0 ~ A_8)	4
	Auto precharge enable										H		4,5
Write & column address	Auto precharge disable		H	X	L	H	L	L	X	V	L	Column address (A_0 ~ A_8)	4
	Auto precharge enable										H		4,5
Burst stop			H	X	L	H	H	L	X	X			6
Precharge	Bank selection		H	X	L	L	H	L	X	V	L	X	
	All banks									X	H		
Clock suspend or active power down		Entry	H	L	H	X	X	X	X	X			
					L	V	V	V					
		Exit	L	H	X	X	X	X	X				
Precharge power down mode		Entry	H	L	H	X	X	X	X	X			
					L	H	H	H					
		Exit	L	H	H	X	X	X	X				
					L	V	V	V					
DQM			H	X					V	X			7
No operation command			H	X	H	X	X	X	X	X			
					L	H	H	H					

<Table 2. 13. Simplified Truth Table>

- 16Mx16x2 SDRAM Linear Bank Addressing Parameters

Parameter	Value	SDCTL0 Register Settings
ROW Address bits	13	ROW = ‘10’
COLUMN Address bits	9	COL = ‘01’
Data Size	32	DSIZ = ‘1x’
Refresh Rate	8192 rows/64 ms	SREFR = ‘11’
Bank Address Mode	Linear Bank Addressing	IAM = ‘0’

<Table 2. 14. Linear Bank Addressing Parameters>

Pin	Name	Type	Configuration	Description
XA[14:0]	MMXA[14..0]	Output		Address BUS (With BA0/1)
XD[31:0]	XD[31..0]	In/Out		Data BUS
SDCLK_X	SDCLK	Output		SDCLK0/ SDCLK1
nSCSx0	MMnSCS0	Output		
nSCSx1	MMnSCS1	Output		
nRAS	nRAS	Output		
nCAS	nCAS	Output		
DQM[3:0]	DQM[3..0]	Output		

<Table 2. 15. SDRAM BUS>

2.8. Touch Panel

By using internal comparator & ADC of MP2520F, it is possible to connect touch panel. For touch panel control, four FETs are necessary.

2.8.1. GPIO for Touch(FDC6321)

Pin	Name	Type	Configuration	Description
GPIOF[3]	TSMY_EN	Output	NONE	Control signal for external FET
GPIOF[4]	TSPY_EN	Output	NONE	Control signal for external FET
GPIOF[5]	TSMX_EN	Output	NONE	Control signal for external FET

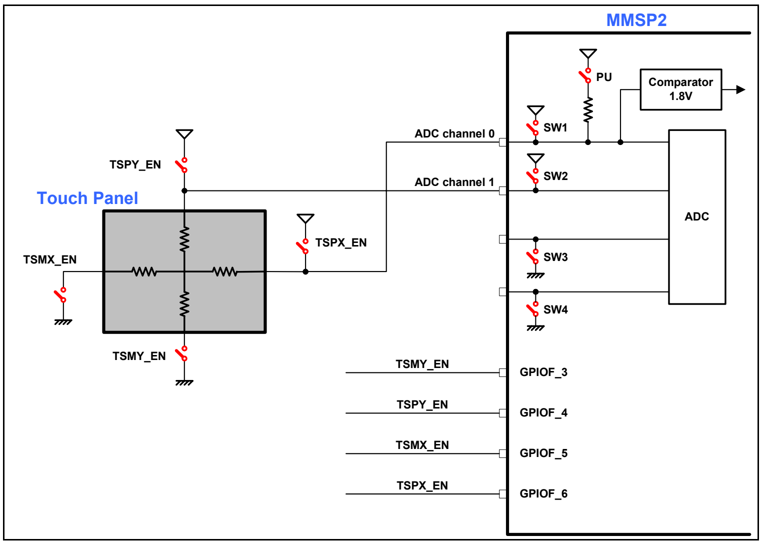
Pin	Name	Type	Configuration	Description
GPIOF[6]	TSPX_EN	Output	NONE	Control signal for external FET

<Table 2. 16. GPIO for Touch>

● External switch : FDC6321

Pin	Name	Type	Configuration	Description
ADCIN0	TSPX		NONE	Y value analog input
ADCIN1	TSPY		NONE	X value analog input

<Table 2. 17. ADC>

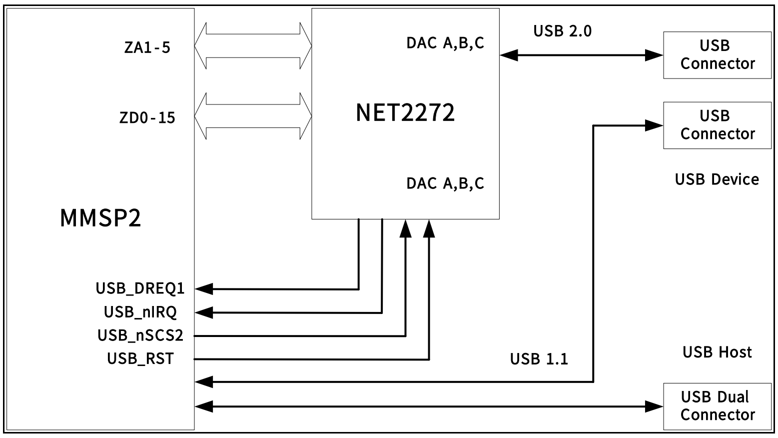


<Fig 2. 5. Touch Panel connection diagram>

2.9. USB Host/Device

● Netchip2272 : USB Device 2.0(1ch)

● MMSP2 : USB Device 1.1(1ch), Host 1.1(2ch)



<Fig 2. 6. USB Block Diagram of Development Board 4.1.1>

2.9.1. USB Device (USB 2.0)

Pin	Net Name	Type	Configuration	Description
GPIOE[15:0]	ZD[15:0]	Alt1	NONE	Data Bus
GPIOG[5:1]	ZA[5:1]	Alt1	NONE	Address Bus
GPIOI[10]	USB_nSCS2	Alt1	NONE	Net2272 Chip select
GPIOD[4]	nUSB_RST	Output	Ext P/U	Net2272 Reset Low Enable
GPIOJ[11]	nPWE	Alt1	NONE	nIOW
GPIOJ[12]	nPOE	Alt1	NONE	nIOR
GPIOK[1]	USB_DACK1	Alt1	NONE	
GPIOK[5]	USB_DREQ1	Alt1	NONE	
GPIOJ[14]	USB_nIRQ	Input	NONE	

<Table 2. 18. GPIO for USB Device(Net2272)>

● Width of the USB Reset : 10ms(minimum)

● USB Reset period may be longer than 10ms depending on the upstream host or hub

● Normal initialization Sequence

- Assert/ De-assert Reset pin
- Local CPU Initializes USB and local bus configuration register

2.9.2. USB Host (USB1.1)

Pin	Net Name	Type	Configuration	Description
GPIOJ[13]	USB_PO_EN	Output	Ext P/U	USB Host Power Enable Low Enable

<Table 2. 19. GPIO for USB Host(MMSP2)>

Address	Register Name	Register Description	Page
00h	REGADDRPTR	Register Address Pointer	56
01h	REGDATA	Register Data	56
02h	IRQSTAT0	Interrupt Status Register (low byte)	56
03h	IRQSTAT1	Interrupt Status Register (high byte)	57
04h	PAGESEL	Endpoint Page Select Register	57
1Ch	DMAREQ	DMA Request Control	58
1Dh	SCRATCH	General Purpose Scratch-pad	58
20h	IRQENB0	Interrupt Enable Register (low byte)	59
21h	IRQENB1	Interrupt Enable Register (high byte)	59
22h	LOCCTL	Local Bus Control	60
23h	CHIPREV_LEGACY	Legacy Chip Silicon Revision	60
24h	LOCCTL1	Local Bus Control 1	60
25h	CHIPREV_2272	Net2272 Chip Silicon Revision	61

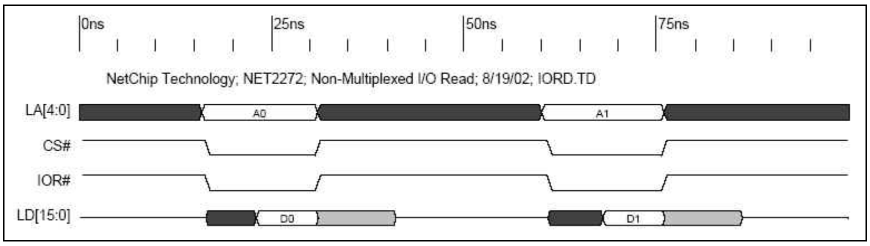
<Table 2. 20. Main Control Register>

Address	Register Name	Register Description	Page
18h	USBCTL0	USB Control Register (low byte)	62
19h	USBCTL1	USB Control Register (high byte)	62
1Ah	FRAME0	USB Frame Number (low byte)	62
1Bh	FRAME1	USB Frame Number (high byte)	62
30h	OURADDR	Our USB Address	63
31h	USBDIAG	Diagnostic register	63
32h	USBTEST	USB 2.0 Test Control Register	64
33h	XCVRDIAG	USB Transceiver Diagnostic Register	64
34h	VIRTOUT0	Virtual OUT Interrupt 0	64
35h	VIRTOUT1	Virtual OUT Interrupt 1	65
36h	VIRTIN0	Virtual IN Interrupt 0	65
37h	VIRTIN1	Virtual IN Interrupt 1	65
40h	SETUP0	Setup byte 0	65
41h	SETUP1	Setup byte 1	66
42h	SETUP2	Setup byte 2	66
43h	SETUP3	Setup byte 3	66
44h	SETUP4	Setup byte 4	66
45h	SETUP5	Setup byte 5	66
46h	SETUP6	Setup byte 6	67
47h	SETUP7	Setup byte 7	67

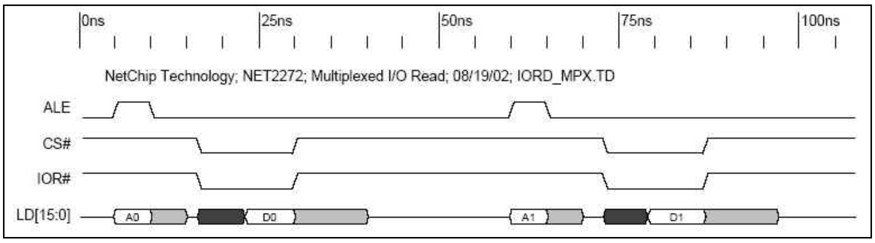
<Table 2. 21. Control Register>

Address	Register Name	Register Description	Page
05h	EP_DATA	Endpoint Data Register	68
06h	EP_STAT0	Endpoint Status (low byte)	68
07h	EP_STAT1	Endpoint Status (high byte)	69
08h	EP_TRANSFER0	IN endpoint byte count (byte 0)	69
09h	EP_TRANSFER1	IN endpoint byte count (byte 1)	69
0Ah	EP_TRANSFER2	IN endpoint byte count (byte 2)	69
0Bh	EP_IRQENB	Endpoint interrupt enable	70
0Ch	EP_AVAIL0	Buffer space/byte count (byte 0)	70
0Dh	EP_AVAIL1	Buffer space/byte count (byte 1)	70
0Eh	EP_RSPCLR	Endpoint Response Control Clear	71
0Fh	EP_RSPSET	Endpoint Response Control Set	72
28h	EP_MAXPKT0	Endpoint Maximum Packet (low byte)	72
29h	EP_MAXPKT1	Endpoint Maximum Packet (high byte)	72
2Ah	EP_CFG	Endpoint configuration	73
2Bh	EP_HBW	Endpoint high bandwidth	73
2Ch	EP_BUFF_STATES	Endpoint buffer states	74

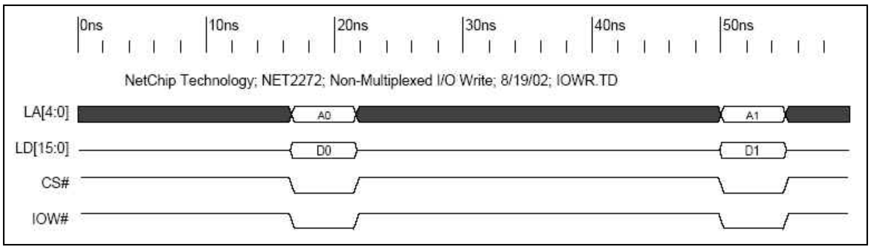
<Table 2. 22. Endpoint Register>



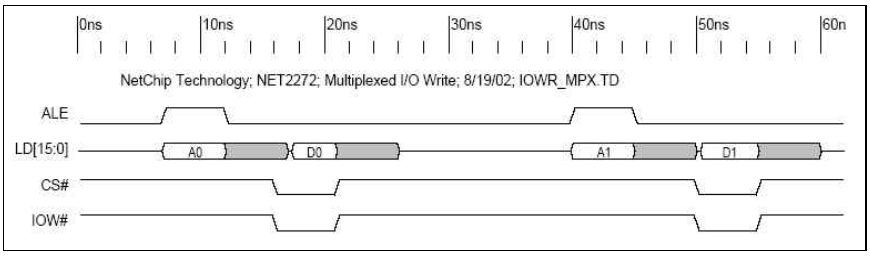
<Fig 2. 7. Non-Multiplexed I/O Read>



<Fig 2. 8. Multiplexed I/O Read>



<Fig 2. 9. Non-Multiplexed I/O Write>



<Fig 2. 10. Multiplexed I/O Write>

Non-Multiplexed I/O Read	Read access Time(T6) : 18nsec Read recovery Time(T8) : 19nsec 8bit bus : 27Mbyte/sec (MAX) 16bit bus : 54Mbyte/sec (MAX)
Multiplexed I/O Read	ALE Width(T3) : 5nsec ALE to READ Command(T19) : 1nsec(minimum) Read access Time(T6) : 18nsec Read recovery Time(T8) : 19nsec 8bit bus : 23Mbyte/sec (1/43nsec) 16bit bus : 46Mbyte/sec (2/43nsec)
Non-Multiplexed I/O Write	Write width(T12) : 5nsec Write to write recovery Time(T15A) : 28nsec 8bit bus : 30Mbyte/sec(1/33nsec) 16bit bus : 60Mbyte/sec(2/33nsec)
Multiplexed I/O Write	ALE width(T3) : 5nsec ALE to WRITE Command(T19) : 1nsec(minimum) Write width(T12) : 5nsec Write to write recovery Time(T15A) : 28nsec 8bit bus : 25Mbyte/sec(1/39nsec) 16bit bus : 50Mbyte/sec(2/39nsec)

<Table 2. 23. I/O Performance for Net2272>

2.10. I2C

● H/W I2C and GPIO I2C can be used for I2C

Function	Pin	Resistor	Description
H/W I2C	SDA(V8), SCL(Y8)	R96, R100	
GPIO I2C	GPIOB[0], BPIOB[1]	R97, R99	

<Table 2. 24. Resistor Setting for I2C>

Pin	Net Name	Type	Configuration	Description
GPIOB[0]	GPIO_SDA	Out	Ext P/U	I2C Data
GPIOB[1]	GPIO_SCL	Out	Ext P/U	I2C Clock

<Table 2. 25. GPIO for I2C>

2.11. Camera

● Controlled by I2C

● CMOS Image sensor : Omnivision, OV9640

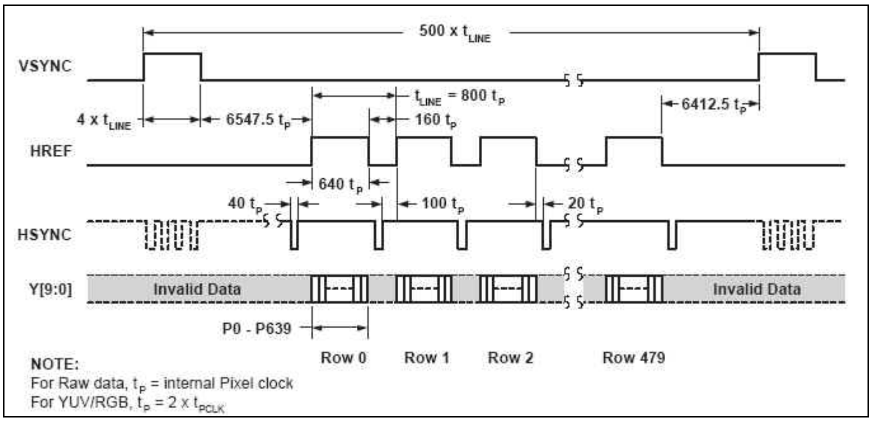
● Camera Module : Nanovision, NVM-O9640

● 24Mhz(ICLK), 640x480x30fps. (Default 15fps)

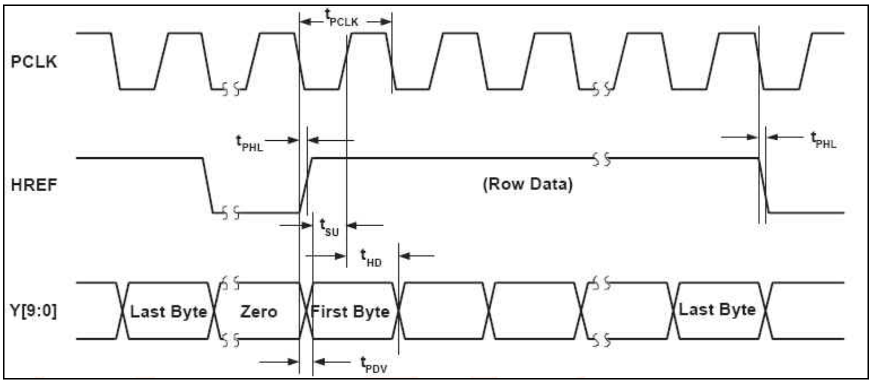
Pin	Name	Type	Configuration	Description
GPIOC[0]	ID0	Alt1	NONE	Data[0]
GPIOC[1]	ID1	Alt1	NONE	Data[1]
GPIOC[2]	ID2	Alt1	NONE	Data[2]
GPIOC[3]	ID3	Alt1	NONE	Data[3]
GPIOC[4]	ID4	Alt1	NONE	Data[4]
GPIOC[5]	ID5	Alt1	NONE	Data[5]
GPIOC[6]	ID6	Alt1	NONE	Data[6]
GPIOC[7]	ID7	Alt1	NONE	Data[7]
GPIOO[2]	CIS_PWDN	Output	NONE	0 : Normal Mode(Default) 1 : Power Down Mode
GPIOO[3]	CIS_Reset	Output	NONE	0 : Normal (Default)

Pin	Name	Type	Configuration	Description
				1 : Reset Enable
GPIOH[1]	CIS_ CLKO	Alt1	NONE	Clock Output
GPIOH[2]	CIS_ VSYNC	Alt1	NONE	Vertical Sync
GPIOH[3]	CIS_ HSYNC	Alt1	NONE	Horizontal Sync
GPIOH[4]	CIS_ CLKI	Alt1	NONE	Clock Intput

<Table 2. 26. GPIO for Camera>



<Fig 2. 11. VGA Frame Timing>



<Fig 2. 12. Horizontal Timing>

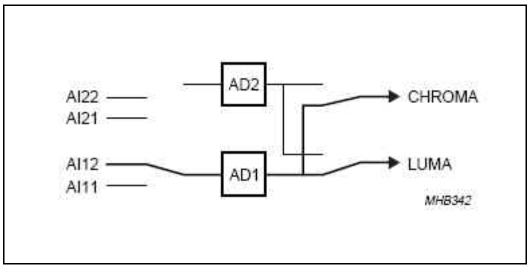
- CIS_PCLK latch data at falling edge
- When CIS_PWDN is Low, Camera Module Power On.
- When CIS_RESET is High, It's reset. OV9640 is remaining low for 1ms and goes to high.

2.12. Video Decoder

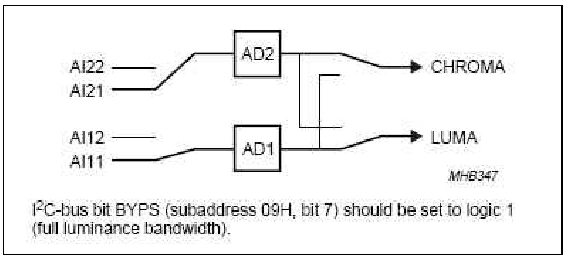
- Philips SAA7113H
- SAA7113H can receive 4 analog inputs
- S-Video/ CVBS in (DTK 4.0 used 3 analog input ports)
- I2C (Address : 48/49H)
- DEC_CE : Decoder Chip Enable

2.12.1. Input

- CVBS : SAA7113H, 7pin (AI21)
- Y(for S-Video) : SAA7113H, 4pin (AI11)
- C(for S-Video) : SAA7113H, 43pin (AI21)
- Refer to SAA7113 Datasheet



<Fig 2. 13. Mode 1 : CVBS (Automatic Gain)>



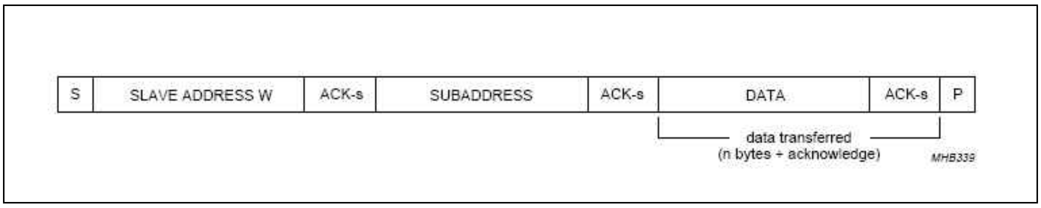
<Fig 2. 14. Mode 8 : Y + C (Gain channel 2 adapted to Y gain)>

FUNCTION(1)	CONTROL BITS D3 TO D0			
	MODE 3	MODE 2	MODE 1	MODE 0
Mode 0: CVBS (automatic gain) from AI11 (pin 4)	0	0	0	0
Mode 1: CVBS (automatic gain) from AI12 (pin 7)	0	0	0	1
Mode 2: CVBS (automatic gain) from AI21 (pin 43)	0	0	1	0
Mode 3: CVBS (automatic gain) from AI22 (pin 1)	0	0	1	1
Mode 4: reserved	0	1	0	0
Mode 5: reserved	0	1	0	1
Mode 6: Y (automatic gain) from AI11 (pin 4) + C (gain adjustable via GAI28 to GAI20) from AI21 (pin 43); note 2	0	1	1	0
Mode 7: Y (automatic gain) from AI12 (pin 7) + C (gain adjustable via GAI28 to GAI20) from AI22 (pin 1); note 2	0	1	1	1
Mode 8: Y (automatic gain) from AI11 (pin 4) + C (gain adapted to Y gain) from AI21 (pin 43); note 2	1	0	0	0
Mode 9: Y (automatic gain) from AI12 (pin 7) + C (gain adapted to Y gain) from AI22 (pin 1); note 2	1	0	0	1
Modes 10 to 15: reserved	1	1	1	1

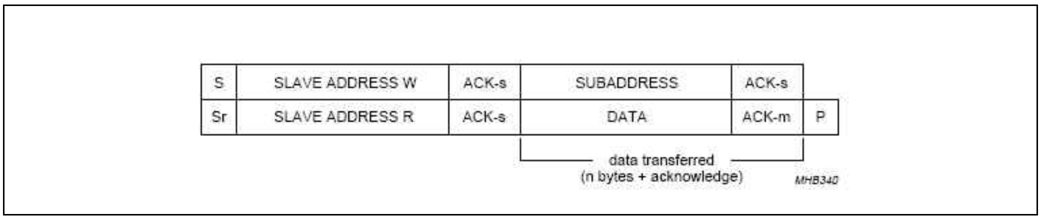
<Table 2. 27. Analog Control 1 SubAddress 01>

<Note>

- Mode Select (See Fig 2.13, 2.14. Please refer to datasheet for more information)
- To take full advantage of the YC-mode, 6 to 9 the I2C bus bit BYPS(Sub address 09H, bit 7) should be set to logic 1(Full luminance bandwidth).



<Fig 2. 15. I2C BUS Write procedure Format>



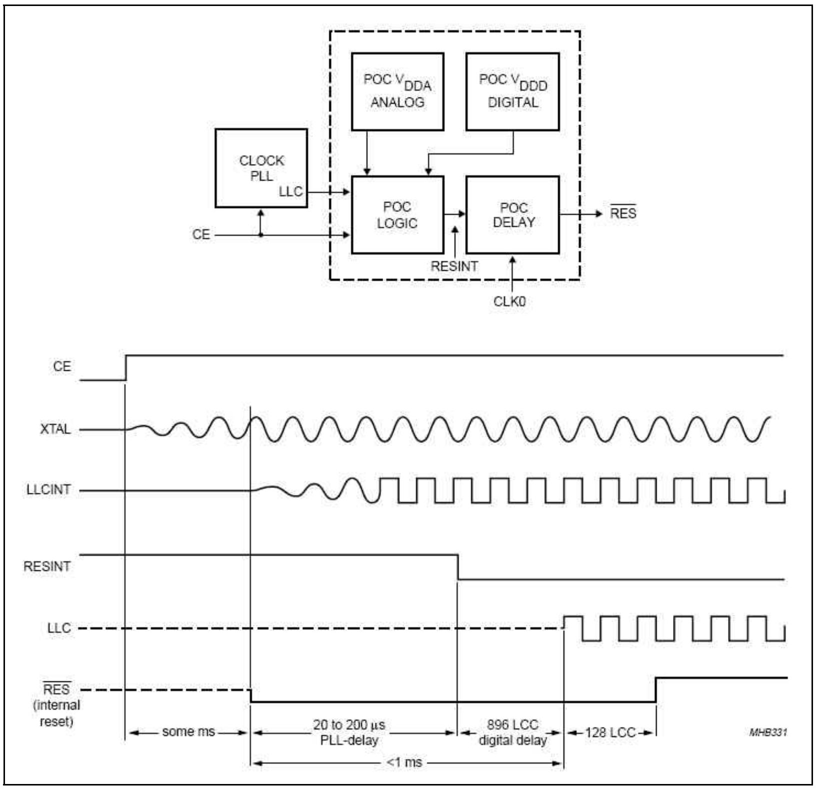
<Fig 2. 16. I2C BUS Read procedure Format>

2.12.2. Output

- CCIR656 format

2.12.3. Power on Reset and Chip Enable Input

- Insufficient digital or analog Vdd supply voltages(below 2.8V) will initiate the reset sequence. All output are force to 3-state.
- It is possible to force a reset by pulling the Chip Enable(CE) to ground.
- After the rising edge of CE and sufficient power supply voltage, the outputs LLC and SDA return from 3-state to active. (Refer to Fig 2.17)

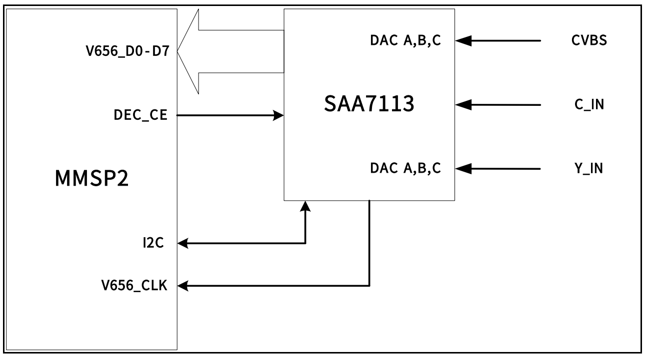


<Fig 2. 17. SAA7113H Initial Sequence>

INTERNAL POWER-ON CONTROL SEQUENCE	PIN OUTPUT STATUS	REMARKS
Directly after power-on asynchronous reset	VPO7 to VPO0, RTCO, RTS0, RTS1, SDA and LLC are in high-impedance state	direct switching to high-impedance for 20 to 200 ms
Synchronous reset sequence	LLC and SDA become active; VPO7 to VPO0, RTCO, RTS0 and RTS1 are held in high-impedance state	internal reset sequence
Status after power-on control sequence	VPO7 to VPO0, RTCO, RTS0 and RTS1 are held in high-impedance state	after power-on (reset sequence) a complete I²C-bus transmission is required

<Table 2. 28. SAA7113H Power-On Control Sequence>

2.12.4. Block Diagram



<Fig 2. 18. Decoder Block Diagram>

2.12.5. GPIO Define

Pin	Name	Type	Configuration	Description
GPIOM[0]	V656_CLK	Alt1	NONE	V656 Clock
GPIOM[1]	V656_D0	Alt1	NONE	Data[0]
GPIOM[2]	V656_D1	Alt1	NONE	Data[1]
GPIOM[3]	V656_D2	Alt1	NONE	Data[2]
GPIOM[4]	V656_D3	Alt1	NONE	Data[3]
GPIOM[5]	V656_D4	Alt1	NONE	Data[4]
GPIOM[6]	V656_D5	Alt1	NONE	Data[5]
GPIOM[7]	V656_D6	Alt1	NONE	Data[6]
GPIOM[8]	V656_D7	Alt1	NONE	Data[7]
GPIOG[15]	DEC_CE	Output	Ext P/U	0 : Chip Disable 1 : Chip Enable (Default)

<Table 2. 29. GPIO of MMSP2 define for SAA7113H>

2.13. Video Encoder

- CX25874 Chip ID : 0x8A
- 4 high-performance, 10bit DACs
- Operation Mode (CX25874)

	Master Mode	Pseudo Maser Mode	Slave Mode
Clock	CX25874 → MMSP2	CX25874 → MMSP2	CX25874 ← MMSP2
H/Vsync	CX25874 → MMSP2	CX25874 ← MMSP2	CX25874 ← MMSP2
Active	CX25874 → MMSP2	CX25874 ← MMSP2	CX25874 ← MMSP2

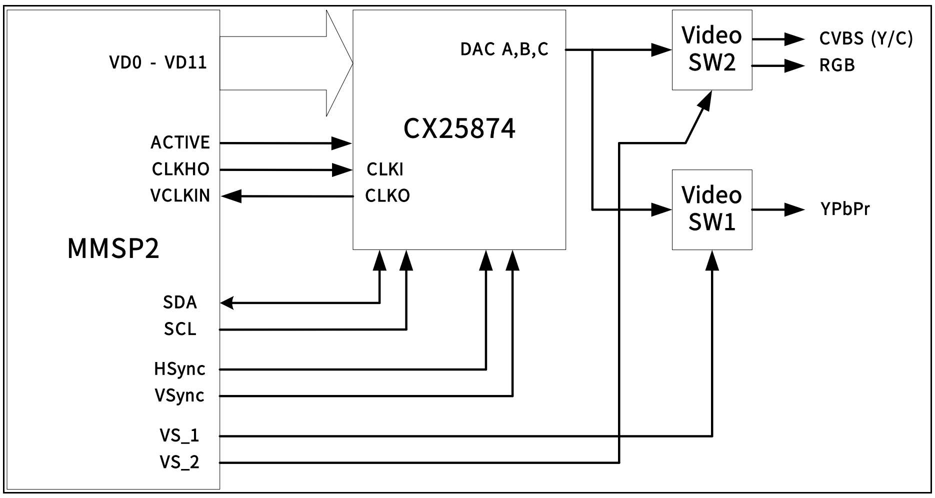
<Table 2. 30. Operation Mode of CX25874>

- DTK4 uses Pseudo Master mode. (MMSP2 does not generate 27MHz clock, so MMSP2 receives a clock from CX25874 and return it to CX25874.). Refer to MP2520F Application Note for more information.
- Video Signal selected by Video Switch 1, 2

VS_1	VS_2	Output	Output Connector
0	X	YPbPr	J14
1	0	CVBS + S-Video	J13
1	1	RGB	J13

<Table 2. 31. Video Switch Selection>

2.13.1. Block Diagram

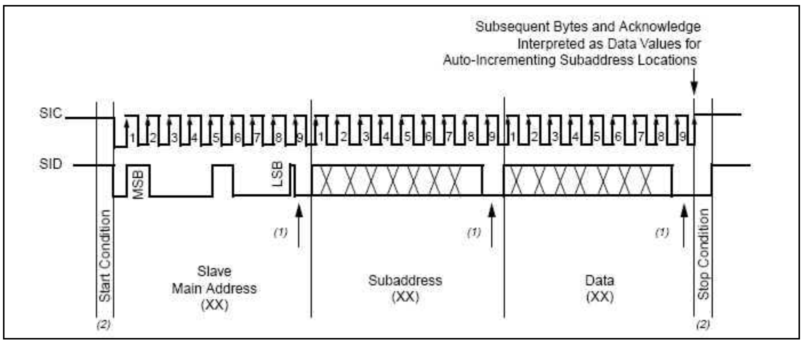


<Fig 2. 19. Encoder Block Diagram>

2.13.2. GPIO Define

Pin	Name	Type	Configuration	Description
GPIOB[8]	VD0	Alt1	NONE	
GPIOB[9]	VD1	Alt1	NONE	
GPIOB[10]	VD2	Alt1	NONE	
GPIOB[11]	VD3	Alt1	NONE	
GPIOB[12]	VD4	Alt1	NONE	
GPIOB[13]	VD5	Alt1	NONE	
GPIOB[14]	VD6	Alt1	NONE	
GPIOB[15]	VD7	Alt1	NONE	
GPIOA[0]	VD8	Alt1	NONE	
GPIOA[1]	VD9	Alt1	NONE	
GPIOA[2]	VD10	Alt1	NONE	
GPIOA[3]	VD11	Alt1	NONE	
GPIOB[2]	CX_PD	Output	Ext P/D	For CX25874 Sleep mode 1 : sleep mode, 0 : normal mode
GPIOB[3]	nCX_RST	Output	Ext P/U	For CX25874 Reset
BPIOB[4]	VSync	Alt1	NONE	
GPIOB[5]	HSync	Alt1	NONE	
GPIOB[6]	Active	Alt1	NONE	Data enable for CX25874
GPIOB[7]	CLKHO	Alt1	NONE	Clock out signal to CX25874
GPIOH[5]	CX_CLKI	Alt1	NONE	Clock in signal from CX25874
GPION[5]	VS_2	Output	NONE	For Video Switch Out enable
GPION[6]	VS_1	Output	NONE	For Video Switch Out enable

<Table 2. 32. GPIO of MMSP2 define for CX25874>



<Fig 2. 20. Programming Diagram for I2C>

2.14. Ethernet

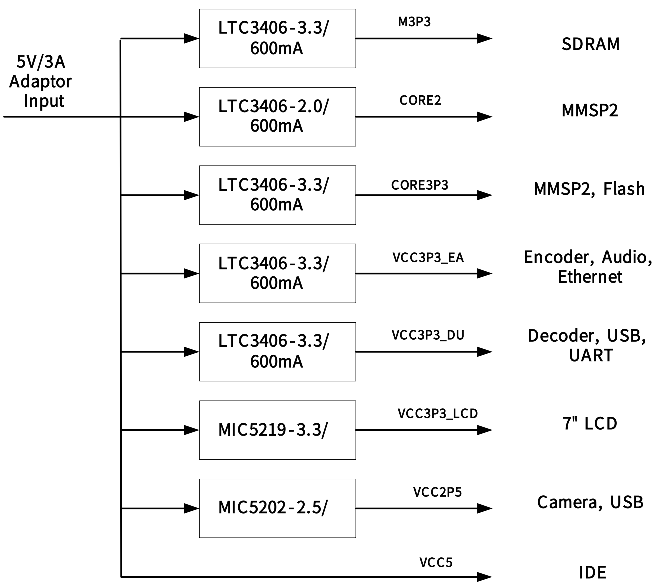
- CS8900
- Init Address : 0x300h
- IO Mode
- Reset : High enable (400ns)

Pin	Net Name	Type	Configuration	Description
GPIOE[15:0]	ZD[15:0]	Alt1	NONE	Data Bus
GPIOG[3:1]	ZA[3:1]	Alt1	NONE	Address Bus
GPIOI[9]	ETHER_nSCS1	Alt1	NONE	Chip select signal for CS8900
GPIOI[12]	ETHER_nBTENB1	Alt1	Ext P/U	Byte enable signal
GPION[4]	ETHER_IRQ	Input	Ext P/D	Interrupt request signal from CS8900
GPIOI[13]	ETHER_IORDY	Input	Ext P/U	Wait signal from CS8900

<Table 2. 33. GPIO for Ethernet>

2.15. Power Block

- **5V/3A Adaptor**
- **2V, 2.5V, 3.3V, 5V**



<Fig 2. 21. Power Block>

2.16. IDE Interface

- **1.8”, 2.5” HDD**
- **5V input**
- **Support PIO and normal DMA(not Ultra DMA or Multiword DMA) mode input/output of the ATA Hard Disk.**
- **Support HDD without nPIOIS16 signal**

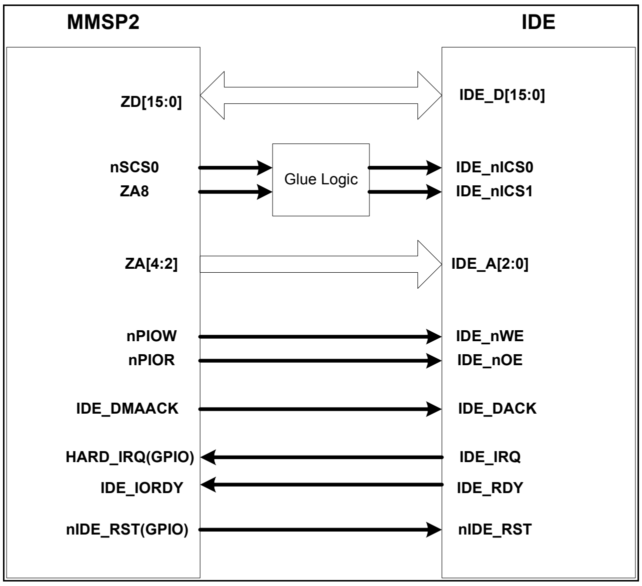
Pin	Net Name	Type	Configuration	Description
GPIOE[15:0]	ZD[15..0]	Alt1	NONE	Data bus
GPIOG[2:0]	ZA[4..2]	Alt1	NONE	Address bus
GPIOI[13]	IDE_IORDY	Alt1	Ext P/U	Wait signal from HDD
GPION[3]	HARD_IRQ	Input	Ext P/D	Interrupt request signal from HDD
GPIOJ[9]	nPWE	Alt1	NONE	Write enable signal for HDD
GPIOJ[10]	nPOE	Alt1	NONE	Read enable signal for HDD
GPION[7]	nIDE_RST	Output	Ext P/U	Reset signal for HDD

<Table 2. 34. GPIO for IDE>

PIN No.	SIGNALS	PIN No.	SIGNALS
1	- RESET	2	GROUND
3	DD 7	4	DD 8
5	DD 6	6	DD 9
7	DD 5	8	DD 10
9	DD 4	10	DD 11
11	DD 3	12	DD 12
13	DD 2	14	DD 13
15	DD 1	16	DD 14
17	DD 0	18	DD 15
19	GROUND	20	KEY
21	DMARQ	22	GROUND
23	- DIOW STOP	24	GROUND
25	-DIOR -DMARDY HSTROBE	26	GROUND
27	IORDY -DMARDY -DSTROBE	28	CSEL
29	-DMACK	30	GROUND
31	INTRQ	32	- IOCS16
33	DA 1	34	- PDIAG· ·CBLID
35	DA 0	36	DA 2
37	- CS0	38	- CS1
39	- DASP	40	GROUND
41	+ 5V (LOGIC)	42	+ 5V (MOTOR)
43	GROUND	44	RESERVED

Note) Symbol (-) in front of signal name shows negative logic.

<Table 2. 35. Signal Pin Assignment>



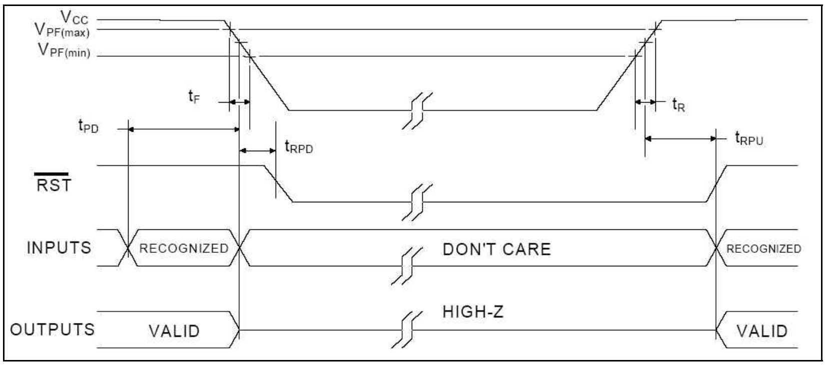
<Fig 2. 22. IDE Block Diagram>

2.17. External RTC

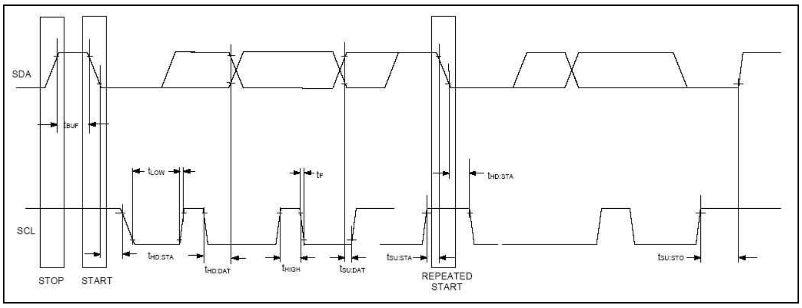
- Using DS1672
- Development Board 4.1.1 includes RTC IC

Pin	Net Name	Type	Configuration	Description
nBATTF	nBATFAULT		Ext P/U	Battery fault signal from external RTC
GPIOB[1]	SCL (GPIO or H/W)	Output	Ext P/U	GPIO I2C Clock signal
SCL			Ext P/U	H/W I2C Clock signal
GPIOB[0]	SDA (GPIO or H/W)	In/Output	Ext P/U	GPIO I2C Data signal
SDA			Ext P/U	H/W I2C Data signal

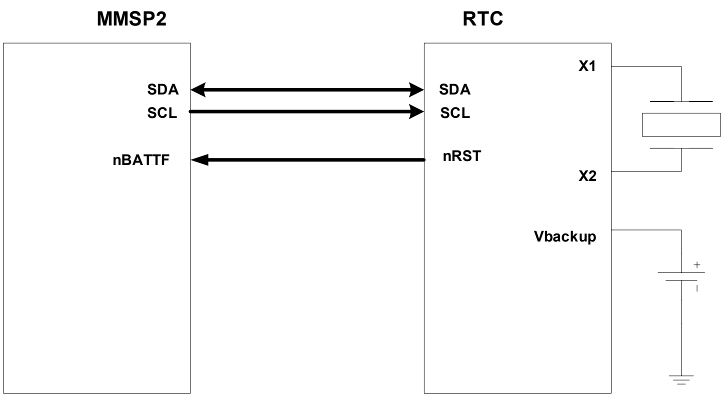
<Table 2. 36. Pin Define for External RTC>



<Fig 2. 23. Power Up/Down Timing>



<Fig 2. 24. Timing Diagram>



<Fig 2. 25. RTC Block Diagram>

3. Board Initial Setting

Pin	Name	Configuration	Description
GPIOA[14]	RstCfgNFBoot	External P/U	**1** : Nand boot Enable 0 : Nand boot Disable
GPIOA[13]	RstCfgNFBSize	External P/D	Nand boot Size **0** : 512byte, 1 : Reserved
GPIOA[12]	RstCfgNFBUSWidth	External P/D	Nand flash bus width **0** : 8bit, 1 : 16bit
GPIOA[11]	RstCfgNFType	External P/U	Nand flash Type 0 : 3address, **1** : 4address
GPIOA[10]	RstCfgSR0BUSWidth	External P/U	SRAM Bus width 0 : 8bit, **1** : 16bit
GPIOA[9]	RstCfgSDRTYPE	External P/D	SDRAM Type **0** : SDRAM, 1: Reserved
GPIOA[8]	RstCfgBUSCBW	External P/U	SDRAM bank bus width **1** : 32bit bus width
GPIOA[7]	RstCfgSDRBW[1]	External P/U	SDRAM Bus Width 00:4bit, 01:8bit, **10**:16bit
GPIOA[6]	RstCfgSDRBW[0]	External P/D	
GPIOA[5]	RstCfgSDRCAP[1]	External P/U	SDRAM Capacity (Mbit) 00 : 64, 01 : 128, **10** : 256, 11 : 512
GPIOA[4]	RstCfgSDRCAP[0]	External P/D	
GPIOA[3]	RstCfgShadow	External P/U	SDRAM Shadow Enable 0 : Disable, **1** : Enable
GPIOA[1]	RstCfgUARTBootClk	External P/D	UART Boot Clk **0** : PLL CLK, 1 : ABITCLK
GPIOA[0]	RstCfgBootDown	External P/D	UART Boot Enable **0** : Disable, 1 : Enable
GPIOB[15]	ARM940T DEBUG Enable	External P/D	ARM940 ICE Debug Enable **0** : Disable, 1 : Enable
GPIOB[14]	RstCfgSRBuf	External P/D	SRAM Buffer Enable **0** : Not use buffer IC 1 : use buffer IC
GPIOB[13]	RstCfgNFBuf	External P/D	Nand Flash Buffer Enable **0** : Not use buffer IC

Pin	Name	Configuration	Description
			1 : use buffer IC

<Table 3. 1. Development Board 4.1.1 initialize configuration>

4. Test Point List

No	Net Name	No	Name	No	Name	No	Name
1	LCLKO	16	nEN_RST	31	FIELD	46	CORE3P3
2	ON_OFF	17		32	XTL_BFO	47	VCC3P3_EA
3	CLKHO	18	nSCS0	33	CVBS1	48	
4	TSMX	19	ETHER_nSCS1	34	VCC2P5	49	
5	TSMY	20	USB_nSCS2	35	CIS_PWDN	50	VCC3P3_LCD
6	nPIOR	21	nBUFOE	36	DEC_CE	51	VCC3P3_DU
7	nPIOW	22	nBUFENB	37	CORE2	52	GPIOO1
8	nNFOE	23	nBTEN	38	GPIOG_14	53	GPIOO0
9	nNFWE	24	ADCIN5	39	GPIOG_13	54	GPIOD3
10	nFRnB	25	ADCIN4	40	GPIOG_12	55	GPION_2
11	nNFCS0	26	nRESET	41	GPIOG_11	56	GPION_1
12	nNFWP	27	nSW_RESET	42	GPIOI_14	57	M3P3
13		28	nPWRRGLTON	43	GPIOI_15	58	
14	VCC5	29		44	LCD_PW	59	
15	GPIOF_2	30	MIC2_IN	45	VCC12	60	

<Table 4. 1. Test Point for Development Board 4.1.1>